

TTIC 31230, Fundamentals of Deep Learning

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Language Modeling

Further Comments on Decoding

We can sample a translation

$$w_t \sim P(w_t \mid \overleftarrow{h}_{\text{in}}[0, J], w_0, \dots, w_{t-1})$$

Typically we do a greedy decoding

$$w_t = \operatorname{argmax}_{w_t} P(w_t \mid \overleftarrow{h}_{\text{in}}[0, J], w_0, \dots, w_{t-1})$$

or we might try maximize total probability.

$$w_0, \dots, w_{T_{\text{out}}} = \operatorname{argmax}_{w_0, \dots, w_{T_{\text{out}}}} P_{\Phi} \left(w_0, \dots, w_{T_{\text{out}}} \mid \overleftarrow{h}_{\text{in}}[0, J] \right)$$

Greedy Decoding vs. Beam Search

We would like

$$W_{\text{out}}[T_{\text{out}}]^* = \operatorname{argmax}_{W_{\text{out}}[T_{\text{out}}]} P_{\Phi}(W_{\text{out}}[T_{\text{out}}] \mid W_{\text{in}}[T_{\text{in}}])$$

But a greedy algorithm may do well

$$w_t = \operatorname{argmax}_{w_t} P_{\Phi}(w_t \mid W_{\text{in}}[T_{\text{in}}], w_0, \dots, w_{t-1})$$

But these are not the same.

Example

“Those apples are good” vs. “Apples are good”

$$P_{\Phi}(\text{Apples are Good } \langle \text{eos} \rangle) > P_{\Phi}(\text{Those apples are good } \langle \text{eos} \rangle)$$

$$P_{\Phi}(\text{Those}|\varepsilon) > P_{\Phi}(\text{Apples}|\varepsilon)$$

Beam Search

At each time step we maintain a list the K best words and their associated hidden vectors.

This can be used to produce a list of k “best” decodings which can then be compared to select the most likely one.

END